

Clustering

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- **Clustering:** Introduction
- Hierarchical Clustering:
 - Agglomerative Clustering Algorithm
 - The single Linkage Algorithm
 - The Complete Linkage Algorithm
 - The Average – Linkage Algorithm
- Partitional Clustering:
 - The K-Means Algorithm

Introduction

- In the earlier chapters, we saw that how samples may be classified if a training set is available to use in the design of a classifier.
- However in many situations classes are themselves are initially undefined.
- Given a set of feature vectors sampled from some population, we would like to know if the data set consists of a number of relatively distinct subsets, then we can define them to be classes.
- This is sometimes called as class discovery or unsupervised classification
- Clustering refers to the process of grouping samples so that the samples are similar within each group. The groups are called clusters.

What is Clustering?

- Clustering is the task of dividing the population or data points into a number of groups such that data points in the same groups are more similar to other data points in the same group than those in other groups.
- In simple words, the aim is to segregate groups with similar traits and assign them into clusters.
- A good clustering will have high intra-class similarity and low inter-class similarity

Applications of Clustering

- Recommendation engines
- Market segmentation
- Social network analysis
- Search result grouping
- Medical imaging
- Image segmentation
- Anomaly detection

Types of clustering:

- Hierarchical Clustering:
 - Agglomerative Clustering Algorithm
 - The single Linkage Algorithm
 - The Complete Linkage Algorithm
 - The Average – Linkage Algorithm
 - Divisive approach
 - Polythetic The division is based on more than one feature.
 - Monothetic Only one feature is considered at a time.
- Partitional Clustering:
 - Forgy's Algorithm
 - The K-Means Algorithm
 - The Isodata Algorithm.

Example: Agglomerative

- 100 students from India join MS program in some particular university in USA.
- Initially each one of them looks like single cluster.
- After some times, 2 students from SJCE, Mysuru makes a cluster.
- Similarly another cluster of 3 students(patterns / Samples) from RVCE meets SJCE students.
- Now these two clusters makes another bigger cluster of Karnataka students.
- Later ... south Indian student cluster and so on...

Example : Divisive approach

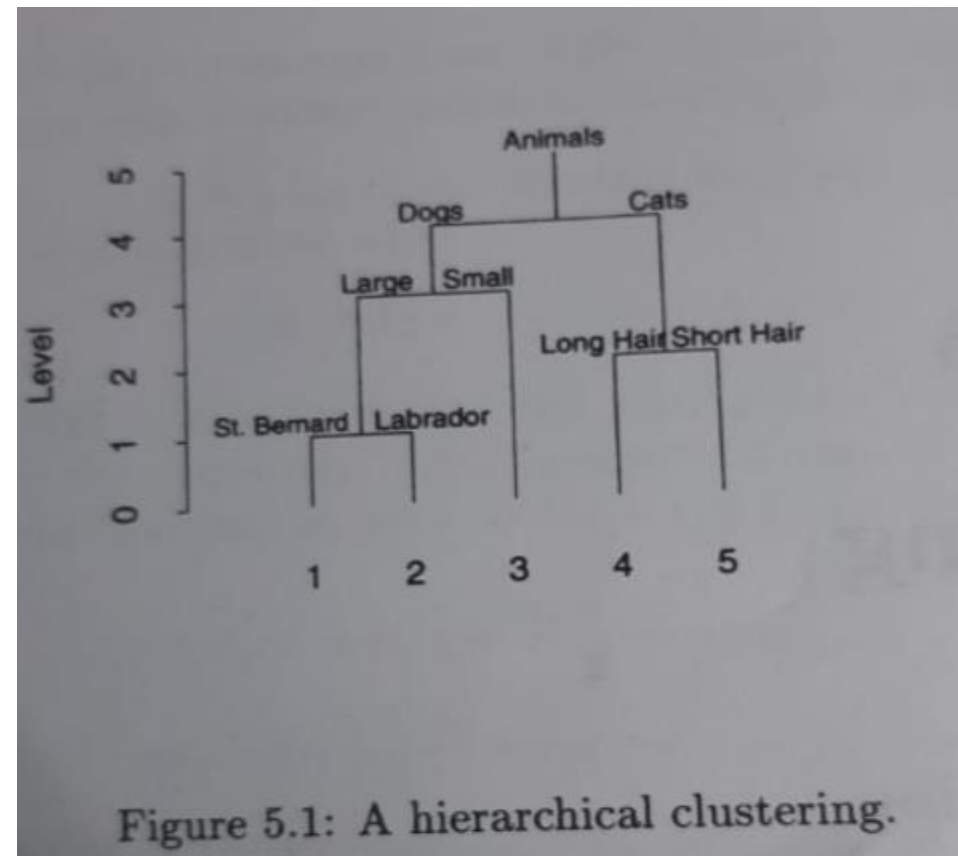
- In a large gathering of engineering students..
 - Separate JSS S&TU students
 - Further computer science students
 - Again ..7th sem students
 - » In sub group and divisive cluster is C section students.

Hierarchical clustering

- Hierarchical clustering refers to a clustering process that organizes the data into large groups, which contain smaller groups and so on.
- A hierarchical clustering may be drawn as a tree or dendrogram.
- The finest grouping is at the bottom of the dendrogram, each sample by itself forms a cluster.
- At the top of the dendrogram, where all samples are grouped into one cluster.

Hierarchical clustering

- Figure shown in figure illustrates hierarchical clustering.
- At the top level we have Animals... followed by sub groups...
- Do not have to assume any particular number of clusters.
- The representation is called dendrogram.
- Any desired number of clusters can be obtained by 'cutting' the **dendrogram** at the proper level.



Two types of Hierarchical Clustering

– Agglomerative:

- It is the most popular algorithm, It is popular than divisive algorithm.
- Start with the points as individual clusters
- It follows bottom up approach
- At each step, merge the closest pair of clusters until only one cluster (or k clusters) left
- Ex: single-linkage, complete-linkage, Average linking algorithm etc.

– Divisive:

- Start with one, all-inclusive cluster
- At each step, split a cluster until each cluster contains a point (or there are k clusters)
- Traditional hierarchical algorithms use a similarity or distance matrix
 - Merge or split one cluster at a time

Agglomerative Clustering Algorithm

1. Compute the proximity matrix
2. Let each data point be a cluster
3. **Repeat**
4. Merge the two closest clusters
5. Update the proximity matrix
6. **Until** only a single cluster remains

Key operation is the computation of the proximity of two clusters

- Different approaches to defining the distance between clusters distinguish the different algorithms

Some commonly used criteria in Agglomerative clustering Algorithms

(The most popular distance measure used is Euclidean distance)

Single Linkage:

Distance between two clusters is the smallest pairwise distance between two observations/nodes, each belonging to different clusters.

Complete Linkage:

Distance between two clusters is the largest pairwise distance between two observations/nodes, each belonging to different clusters.

Mean or average linkage clustering:

Distance between two clusters is the average of all the pairwise distances, each node/observation belonging to different clusters.

Centroid linkage clustering:

Distance between two clusters is the distance between their centroids.

Single linkage algorithm

- Consider the following scatter plot points.
- In single link hierarchical clustering, we merge in each step the two clusters, whose two closest members have the smallest distance

	x	y
1	4	4
2	8	4
3	15	8
4	24	4
5	24	12

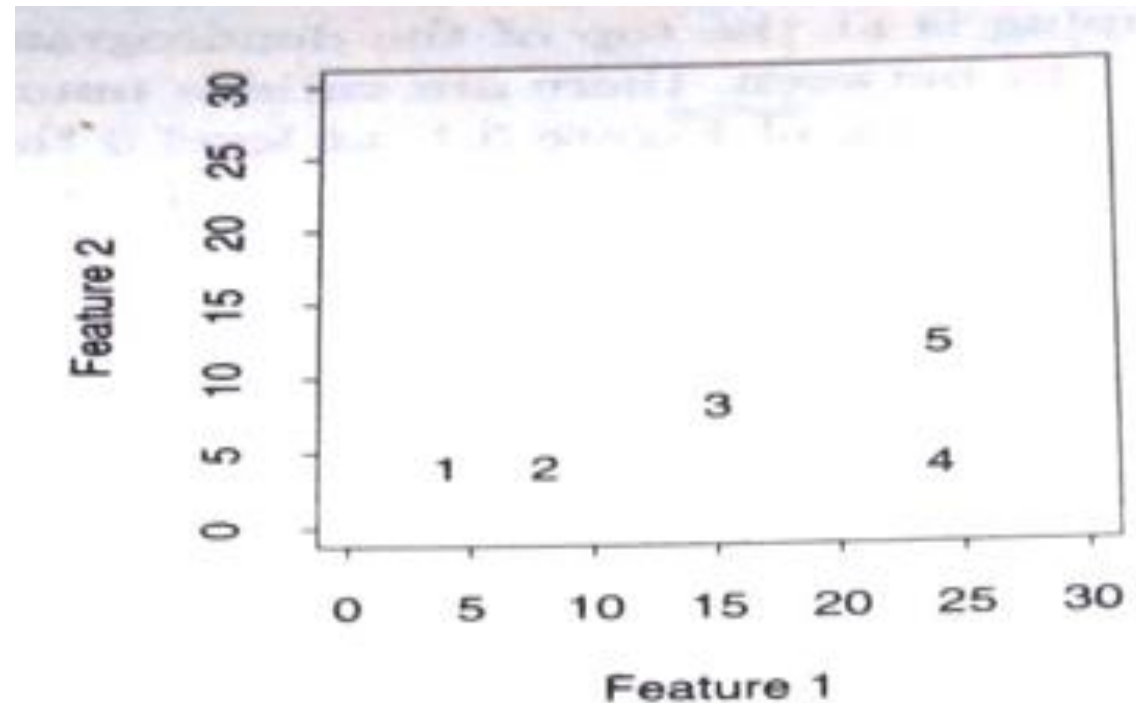


Figure 5.2: Samples for clustering.

Single linkage... Continued

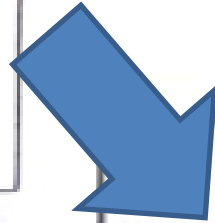
- The single linkage algorithm is also known as the minimum method and the nearest neighbor method.
- Consider C_i and C_j are two clusters.
- 'a' and 'b' are samples from cluster C_i and C_j respectively.

$$D_{SL}(C_i, C_j) = \min_{a \in C_i, b \in C_j} d(a, b),$$

- Where $d(a, b)$ represents the distance between 'a' and 'b'.

First level of distance computation D1 (Euclidean distance used)

	x	y
1	4	4
2	8	4
3	15	8
4	24	4
5	24	12

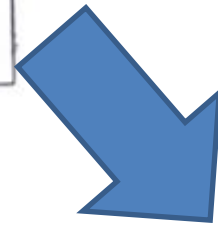


	1	2	3	4	5
1	—	4.0	11.7	20.0	21.5
2	4.0	—	8.1	16.0	17.9
3	11.7	8.1	—	9.8	9.8
4	20.0	16.0	9.8	—	8.0
5	21.5	17.9	9.8	8.0	—

- Use Euclidean distance for distance between samples.
- The table shown in the previous slide gives feature values for each sample and the distance d between each pair of samples.
- The algorithm begins with five clusters, each consisting of one sample.
- The two nearest clusters are then merged.
- The smallest number is 4 which is the distance between (1 and 2), so they are merged. Merged matrix is as shown in next slide.

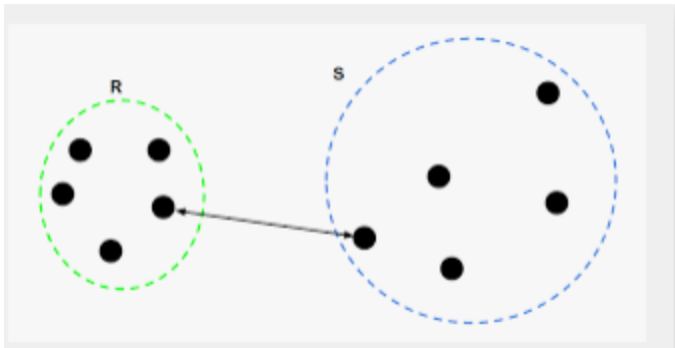
D2 matrix

	1	2	3	4	5
1	—	4.0	11.7	20.0	21.5
2	4.0	—	8.1	16.0	17.9
3	11.7	8.1	—	9.8	9.8
4	20.0	16.0	9.8	—	8.0
5	21.5	17.9	9.8	8.0	—



$\{1, 2\}, \{3\}, \{4\}, \{5\}.$

Next obtain the matrix that gives the distances between these clusters:



	$\{1,2\}$	3	4	5
$\{1,2\}$	—	8.1	16.0	17.9
3	8.1	—	9.8	9.8
4	16.0	9.8	—	8.0
5	17.9	9.8	8.0	—

- In the next level, the smallest number in the matrix is 8
- It is between 4 and 5.
- Now the cluster 4 and 5 are merged.
- With this we will have 3 clusters: $\{1,2\}$, $\{3\}$, $\{4,5\}$
- The matrix is as shown in the next slide.

D3 distance

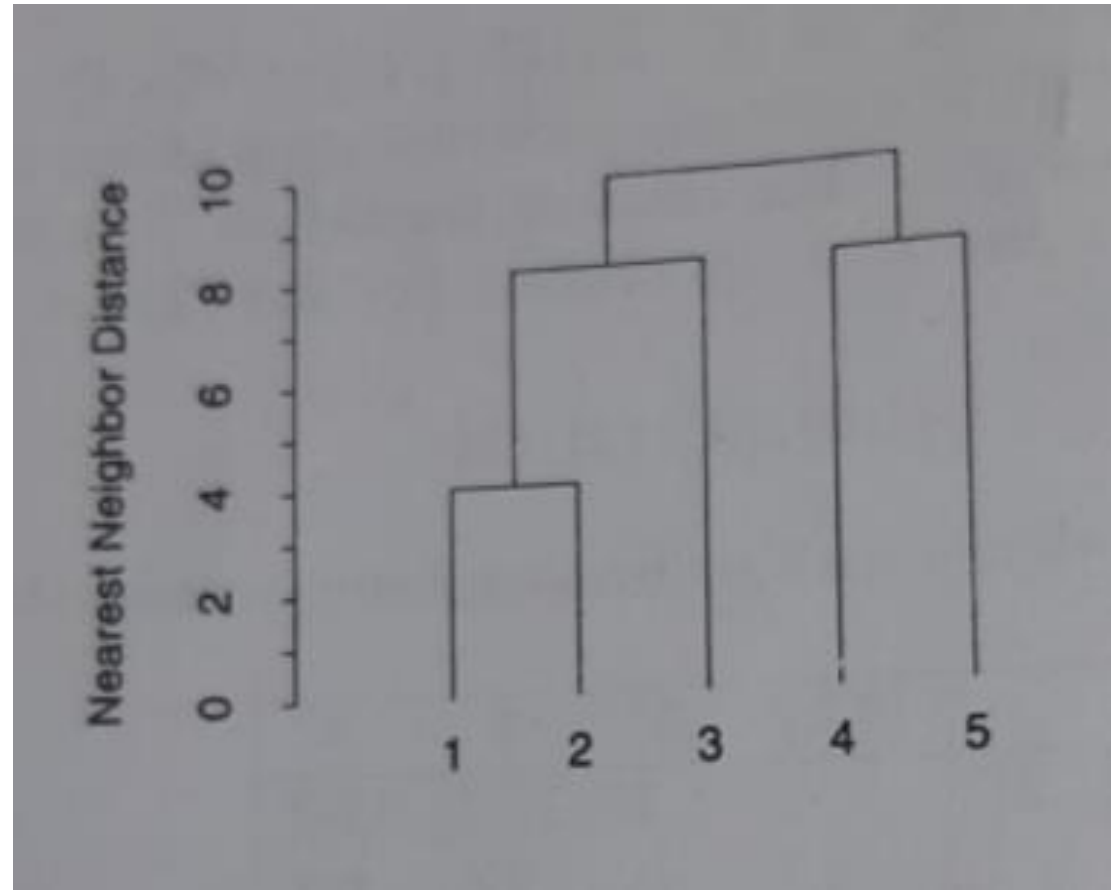
	{1,2}	3	4	5
{1,2}	—	8.1	16.0	17.9
3	8.1	—	9.8	9.8
4	16.0	9.8	—	8.0
5	17.9	9.8	8.0	—



	{1,2}	3	{4,5}
{1,2}	—	8.1	16.0
3	8.1	—	9.8
{4,5}	16.0	9.8	—

- In the next step $\{1,2\}$ will be merged with $\{3\}$.
- Now we will have two cluster $\{1,2,3\}$ and $\{4,5\}$
- In the next step.. these two are merged to have single cluster.
- Dendrogram is as shown here.
- Height of the dendrogram is decided based on the merger distance.

For example: 1 and 2 are merged at the least distance 4. hence the height is 4.



The complete linkage Algorithm

- It is also called the **maximum method or the farthest neighbor method**.
- It is obtained by defining the distance between two clusters to be largest distance between a sample in one cluster and a sample in the other cluster.
- If C_i and C_j are clusters, we define:

$$D_{CL}(C_i, C_j) = \max_{a \in C_i, b \in C_j} d(a, b).$$

Example : Complete linkage algorithm

- Consider the same samples used in single linkage:
- Apply Euclidean distance and compute the distance.
- Algorithm starts with 5 clusters.
- As earlier samples 1 and 2 are the closest, they are merged first.

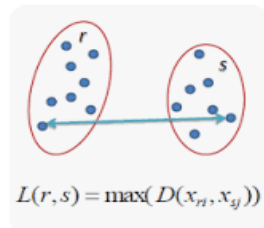
	x	y
1	4	4
2	8	4
3	15	8
4	24	4
5	24	12



	1	2	3	4	5
1	—	4.0	11.7	20.0	21.5
2	4.0	—	8.1	16.0	17.9
3	11.7	8.1	—	9.8	9.8
4	20.0	16.0	9.8	—	8.0
5	21.5	17.9	9.8	8.0	—

- While merging the maximum distance will be used to replace the distance/ cost value.
- For example, the distance between 1&3 = 11.7 and 2&3=8.1. This algorithm selects 11.7 as the distance.
- In complete linkage hierarchical clustering, the distance between two clusters is defined as **the longest distance between two points in each cluster.**

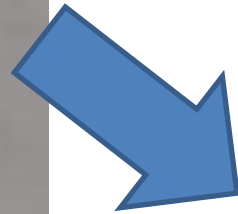
	1	2	3	4	5
1	—	4.0	11.7	20.0	21.5
2	4.0	—	8.1	16.0	17.9
3	11.7	8.1	—	9.8	9.8
4	20.0	16.0	9.8	—	8.0
5	21.5	17.9	9.8	8.0	—



	{1,2}	3	4	5
{1,2}	—	11.7	20.0	21.5
3	11.7	—	9.8	9.8
4	20.0	9.8	—	8.0
5	21.5	9.8	8.0	—

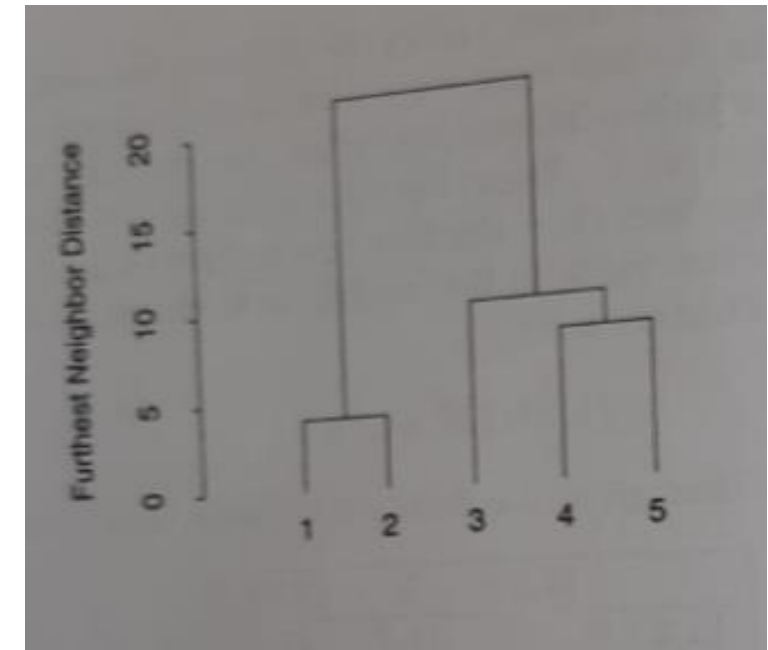
- In the next level, the smallest distance in the matrix is 8.0 between 4 and 5. Now merge 4 and 5.

	{1,2}	3	4	5
{1,2}	—	11.7	20.0	21.5
3	11.7	—	9.8	9.8
4	20.0	9.8	—	8.0
5	21.5	9.8	8.0	—



	{1,2}	3	{4,5}
{1,2}	—	11.7	21.5
3	11.7	—	9.8
{4,5}	21.5	9.8	—

- In the next step, the smallest distance is 9.8 between 3 and {4,5}, they are merged.
- At this stage we will have two clusters {1,2} and {3,4,5}.
- Notice that these clusters are different from those obtained from single linkage algorithm.
- At the next step, the two remaining clusters will be merged.
- The hierarchical clustering will be complete.
- The dendrogram is as shown in the figure.



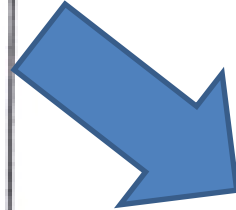
The Average Linkage Algorithm

- The average linkage algorithm, is an attempt to compromise between the extremes of the single and complete linkage algorithm.
- It is also known as the **unweighted pair group method using arithmetic averages**.

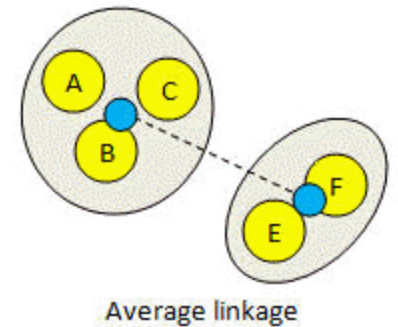
Example: Average linkage clustering algorithm

- Consider the same samples: compute the Euclidian distance between the samples

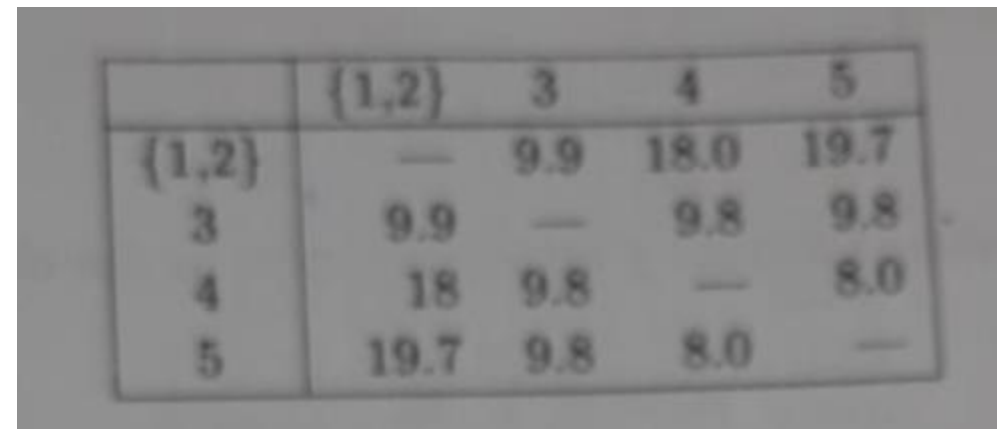
	x	y
1	4	4
2	8	4
3	15	8
4	24	4
5	24	12



	1	2	3	4	5
1	—	4.0	11.7	20.0	21.5
2	4.0	—	8.1	16.0	17.9
3	11.7	8.1	—	9.8	9.8
4	20.0	16.0	9.8	—	8.0
5	21.5	17.9	9.8	8.0	—



- In the next step, cluster 1 and 2 are merged, as the distance between them is the least.
- The distance values are computed based on the average values.
- For example distance between 1 & 3 =11.7 and 2&3=8.1 and the average is 9.9. This value is replaced in the matrix between {1,2} and 3.



A handwritten distance matrix table showing the merging of clusters 1 and 2 into {1,2}. The table has 5 rows and 5 columns. The first row and column are headers for the clusters: {1,2}, 3, 4, and 5. The diagonal elements are dashes, indicating a distance of 0. The values in the matrix represent the distances between the clusters. The value 9.9 is shown in the cells for {1,2} to 3 and 3 to {1,2}, representing the average distance after merging clusters 1 and 2.

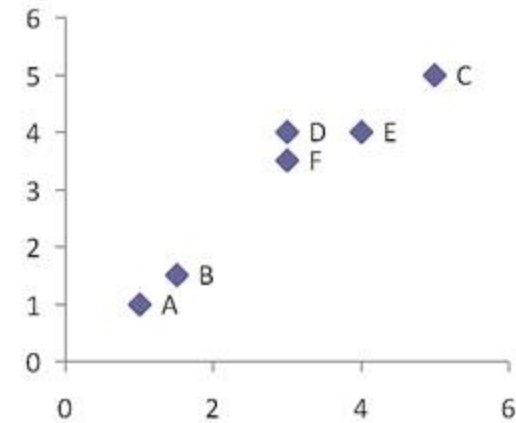
	{1,2}	3	4	5
{1,2}	—	9.9	18.0	19.7
3	9.9	—	9.8	9.8
4	18	9.8	—	8.0
5	19.7	9.8	8.0	—

- In the next stage 4 and 5 are merged:

	$\{1,2\}$	3	$\{4,5\}$
$\{1,2\}$	—	9.9	18.9
3	9.9	—	9.8
$\{4,5\}$	18.9	9.8	—

Example 2: Single Linkage

	X1	X2
A	1	1
B	1.5	1.5
C	5	5
D	3	4
E	4	4
F	3	3.5



Dist	A	B	C	D	E	F
A	0.00	0.71	5.66	3.61	4.24	3.20
B	0.71	0.00	4.95	2.92	3.54	2.50
C	5.66	4.95	0.00	2.24	1.41	2.50
D	3.61	2.92	2.24	0.00	1.00	0.50
E	4.24	3.54	1.41	1.00	0.00	1.12
F	3.20	2.50	2.50	0.50	1.12	0.00

Min Distance (Single Linkage)

Dist	A	B	C	D, F	E
A	0.00	0.71	5.66	?	4.24
B	0.71	0.00	4.95	?	3.54
C	5.66	4.95	0.00	?	1.41
D, F	?	?	?	0.00	?
E	4.24	3.54	1.41	?	0.00

Then, the updated distance matrix becomes

Min Distance (Single Linkage)

Dist	A	B	C	D, F	E
A	0.00	0.71	5.66	3.20	4.24
B	0.71	0.00	4.95	2.50	3.54
C	5.66	4.95	0.00	2.24	1.41
D, F	3.20	2.50	2.24	0.00	1.00
E	4.24	3.54	1.41	1.00	0.00

Then the updated distance matrix is

Min Distance (Single Linkage)

Dist	A,B	C	(D, F)	E
A,B	0	4.95	2.50	3.54
C	4.95	0	2.24	1.41
(D, F)	2.50	2.24	0	1.00
E	3.54	1.41	1.00	0

Min Distance (Single Linkage)

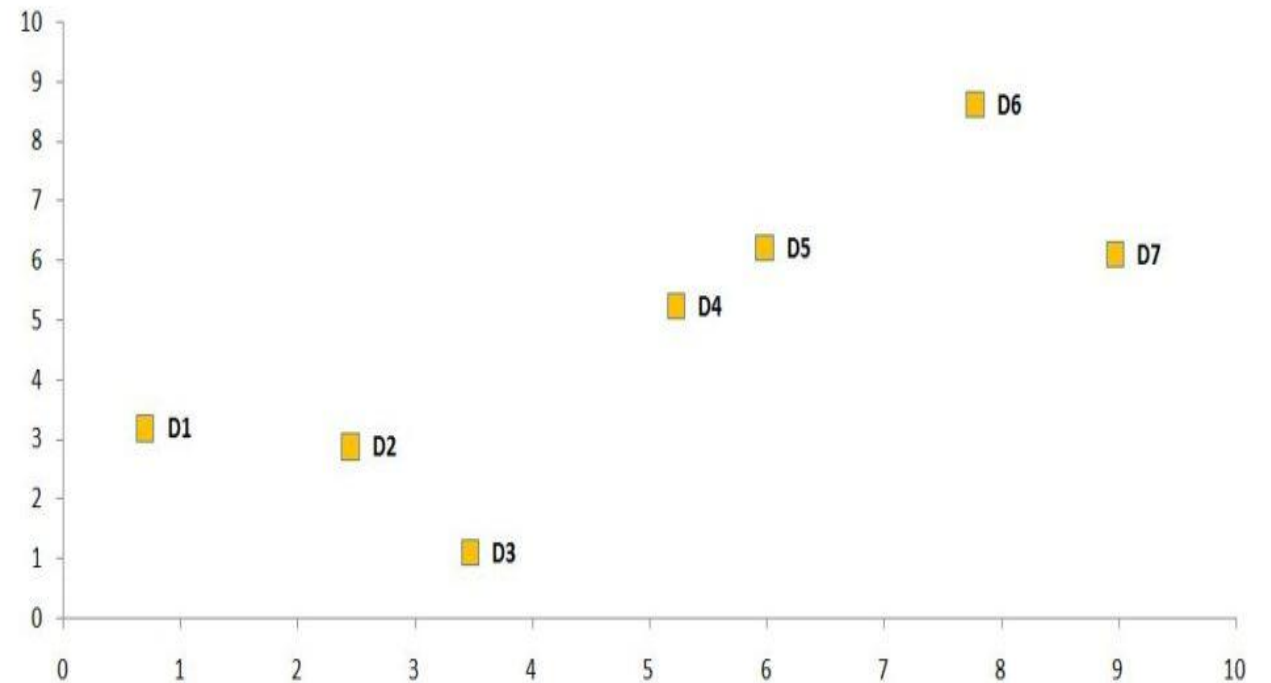
Dist	(A,B)	C	(D, F), E
(A,B)	0.00	4.95	2.50
C	4.95	0.00	1.41
(D, F), E	2.50	1.41	0.00

Min Distance (Single Linkage)

Dist	(A,B)	((D, F), E), C
(A,B)	0.00	2.50
((D, F), E), C	2.50	0.00

Example 3: Single linkage

Data Points	X	Y
D1	0.7	3.2
D2	2.45	2.89
D3	3.47	1.12
D4	5.23	5.24
D5	5.98	6.23
D6	7.778	8.63
D7	8.97	6.12

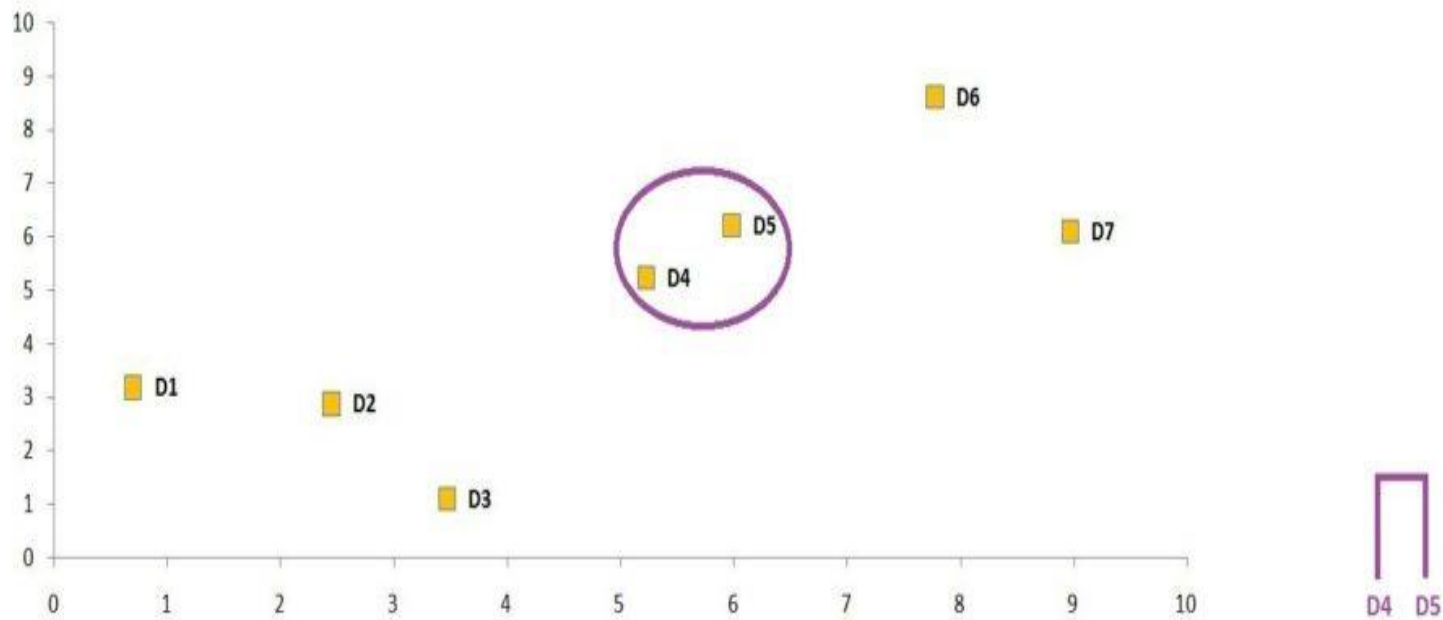


	D1	D2	D3	D4	D5	D6	D7
D1	0	1.78	3.46	4.97	6.09	8.92	8.77
D2	1.78	0	2.04	3.64	4.86	7.83	7.28
D3	3.46	2.04	0	4.48	5.69	8.66	7.43
D4	4.97	3.64	4.48	0	1.24	4.24	3.84
D5	6.09	4.86	5.69	1.24	0	3.00	2.99
D6	8.92	7.83	8.66	4.24	3.00	0	2.78
D7	8.77	7.28	7.43	3.84	2.99	2.78	0

	D1	D2	D3	D4	D5	D6	D7
D1	0	1.78	3.46	4.97	6.09	8.92	8.77
D2	1.78	0	2.04	3.64	4.86	7.83	7.28
D3	3.46	2.04	0	4.48	5.69	8.66	7.43
D4	4.97	3.64	4.48	0	1.24	4.24	3.84
D5	6.09	4.86	5.69	1.24	0	3.00	2.99
D6	8.92	7.83	8.66	4.24	3.00	0	2.78
D7	8.77	7.28	7.43	3.84	2.99	2.78	0

	D1	D2	D3	D4	D5	D6	D7
D1							
D2	1.78						
D3	3.46	2.04					
D4	4.97	3.64	4.48				
D5	6.09	4.86	5.69	1.24			
D6	8.92	7.83	8.66	4.24	3.00		
D7	8.77	7.28	7.43	3.84	2.99	2.78	

	D1	D2	D3	D4	D5	D6	D7
D1							
D2	1.78						
D3	3.46	2.04					
D4	4.97	3.64	4.48				
D5	6.09	4.86	5.69	1.24			
D6	8.92	7.83	8.66	4.24	3.00		
D7	8.77	7.28	7.43	3.84	2.99	2.78	

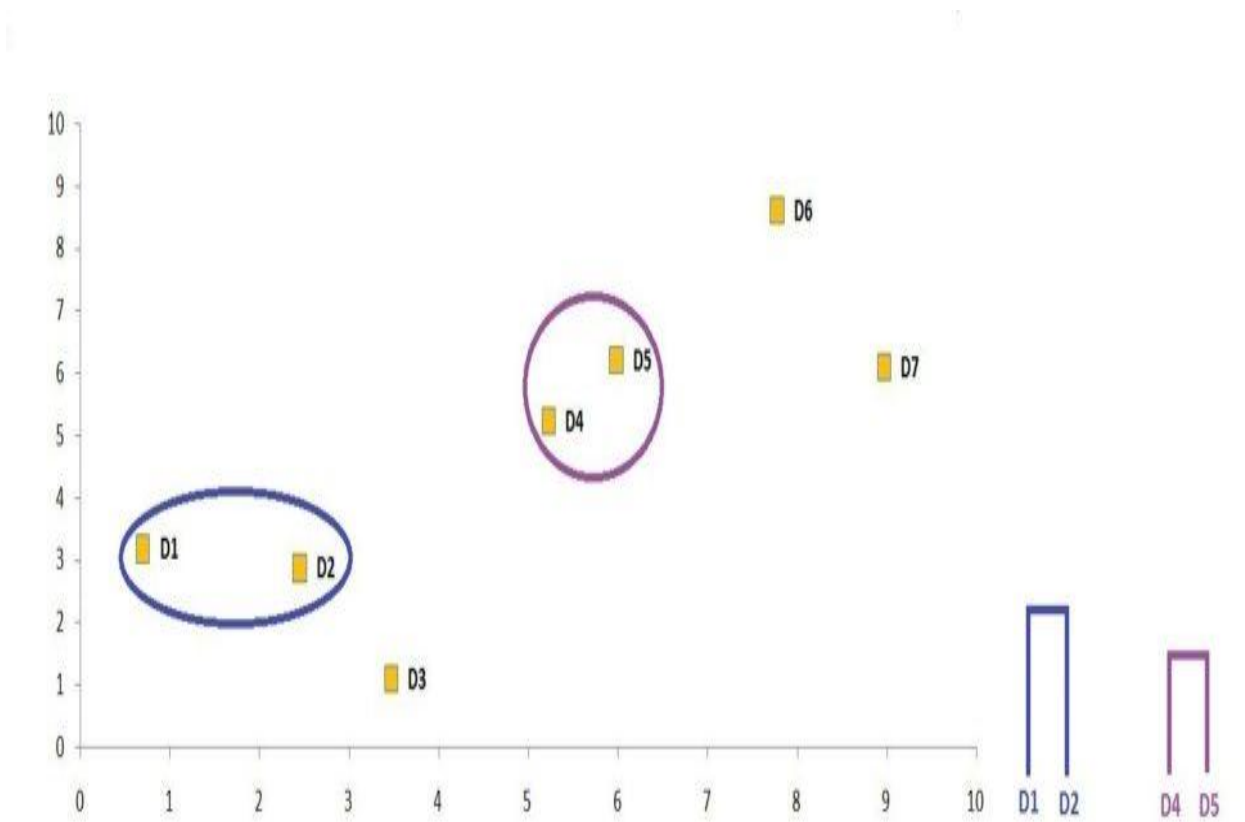


	D1	D2	D3	D4,D5	D6	D7
D1						
D2	1.78					
D3	3.46	2.04				
D4,D5	4.97	3.64	4.48			
D6	8.92	7.83	8.66	3.00		
D7	8.77	7.28	7.43	2.99	2.78	

As we are using single linkage, we choose the minimum distance, therefore, we choose 4.97 and consider it as the distance between the D1 and D4, D5. If we were using complete linkage then the maximum value would have been selected as the distance between D1 and D4, D5 which would have been 6.09. If we were to use Average Linkage then the average of these two distances would have been taken. Thus, here the distance between D1 and D4, D5 would have come out to be 5.53 $(4.97 + 6.09 / 2)$.

From now on we will simply repeat Step 2 and Step 3 until we are left with one cluster. We again look for the minimum value which comes out to be 1.78 indicating that the new cluster which can be formed is by merging the data points D1 and D2.

	D1	D2	D3	D4,D5	D6	D7
D1						
D2	1.78					
D3	3.46	2.04				
D4,D5	4.97	3.64	4.48			
D6	8.92	7.83	8.66	3.00		
D7	8.77	7.28	7.43	2.99	2.78	

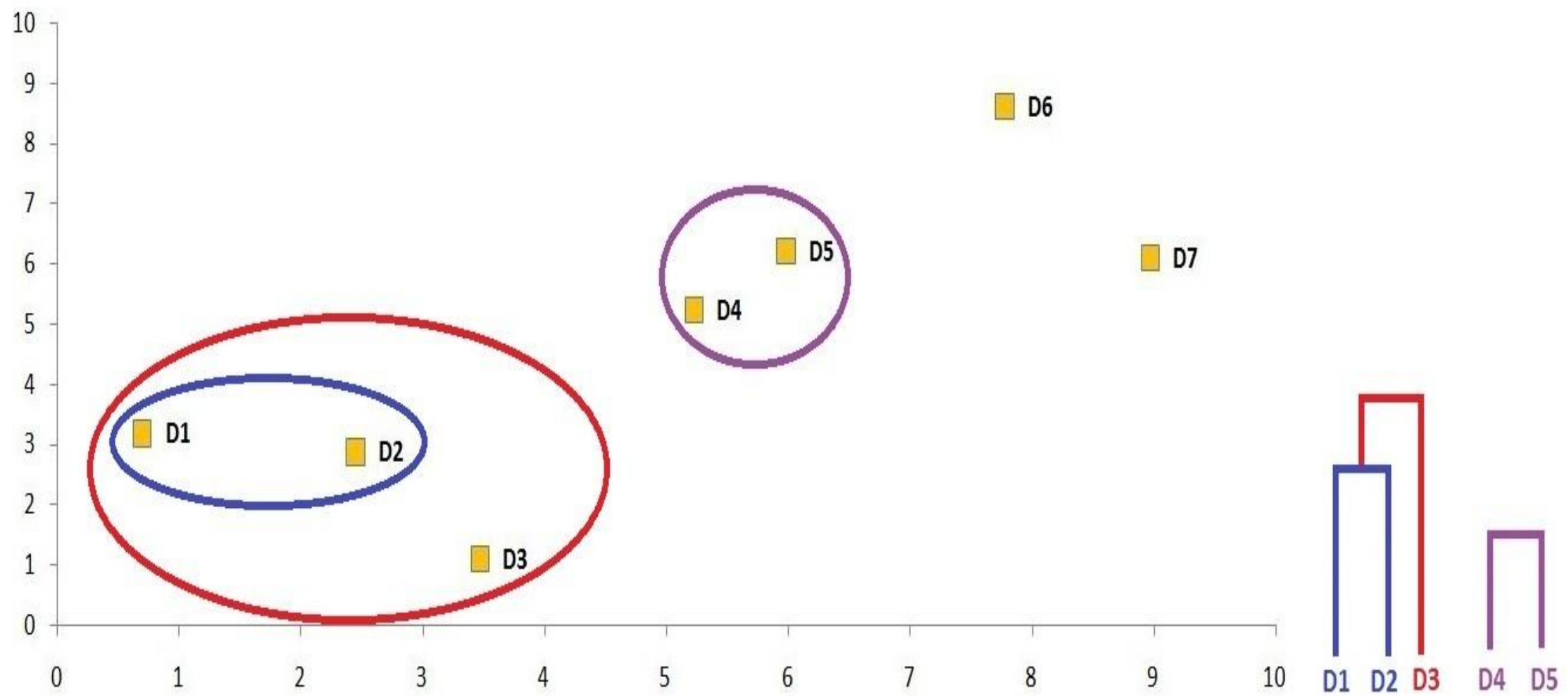


	D1,D2	D3	D4,D5	D6	D7
D1,D2					
D3	1.78				
D4,D5	3.64	4.48			
D6	7.83	8.66	3.00		
D7	7.28	7.43	2.99	2.78	

Similar to what we did in Step 3, we again recalculate the distance this time for cluster D1, D2 and come up with the following updated distance matrix.

	D1,D2	D3	D4,D5	D6	D7
D1,D2					
D3	1.78				
D4,D5	3.64	4.48			
D6	7.83	8.66	3.00		
D7	7.28	7.43	2.99	2.78	

We repeat what we did in step 2 and find the minimum value available in our distance matrix. The minimum value comes out to be 1.78 which indicates that we have to merge D3 to the cluster D1, D2.

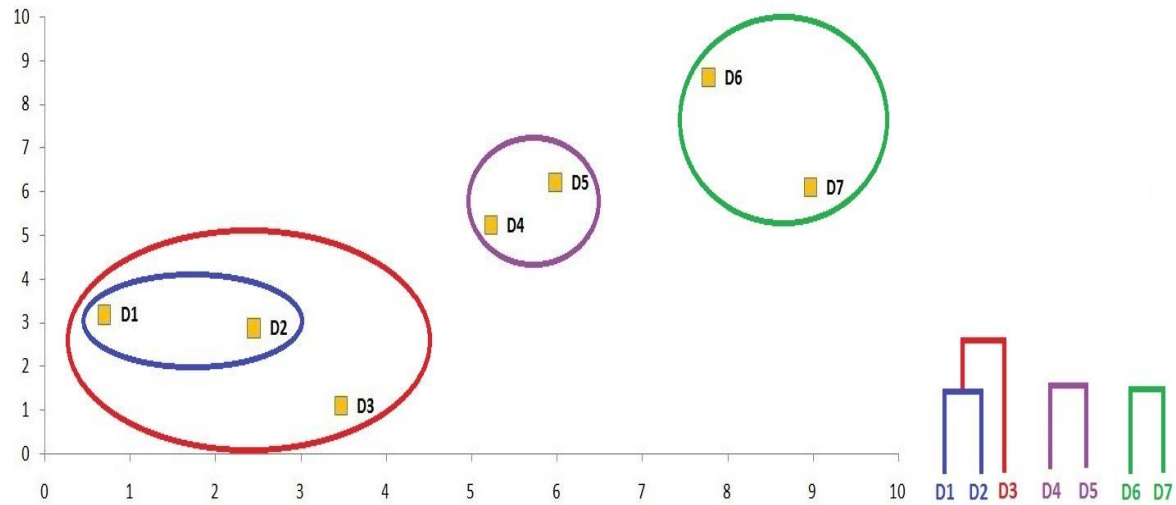


	D1,D2,D3	D4,D5	D6	D7
D1,D2,D3				
D4,D5	3.64			
D6	7.83	3.00		
D7	7.28	2.99	2.78	

Update the distance matrix using
Single Link method.

	D1,D2,D3	D4,D5	D6	D7
D1,D2,D3				
D4,D5	3.64			
D6	7.83	3.00		
D7	7.28	2.99	2.78	

Find the minimum distance in the matrix.



Merge the data points accordingly and form another cluster.

	D1,D2,D3	D4,D5	D6,D7
D1,D2,D3			
D4,D5	3.64		
D6,D7	7.28	2.99	

	D1,D2,D3	D4,D5	D6,D7
D1,D2,D3			
D4,D5	3.64		
D6,D7	7.28	2.99	

Update the distance matrix using Single Link method.